

NATURAL DRAFT FIRED HEATER VER 1

Operator Training Simulator · H-100

Chulalongkorn University · Department of Chemical Engineering

DOCUMENT NO. : NDFH-V1-IN-CE-001

DOCUMENT TITLE : CAUSE AND EFFECT DIAGRAM

Teaching document generated from the simulator logic (bmsTrip, tripConditions, alarm list).

Rev.	Date	Revision status	Prepared	Checked	Approved	Remark
R0	03-Oct-26	Issued for teaching	A. Soottitantawat	—	—	

NATURAL DRAFT FIRED HEATER VER 1 · H-100 CAUSE and EFFECT DIAGRAM Chulalongkorn University · Department of Chemical Engineering								Client :		Chulalongkorn University · Department of Chemical Engineering							REV	DATE	ORIGINATOR	CHECKER	APPROVER	
								PROJECT :		Natural Draft Fired Heater Ver 1 · OTS							R0	03-Oct-26	A. Soottitantawat	—	—	
								DOCUMENT No. :		NDFH-V1-IN-CE-001												
								DOCUMENT TITLE :		Cause and Effect Diagram												
								P&ID :		P&ID-001 rev C / NDFH-V1-PID-001 rev D											SHEET 1 of 1	
Causes								EFFECTS														
								XV-301 · Fuel shut-off valve (upstream)	XV-302 · Fuel shut-off valve (downstream)	XV-303 · Bleed between XV-301/302	FV-300 · Fuel control valve (TIC-030 → MAN 0 %)	FV-200 · Combustion-air valve (FIC-005 → MAN 100 %)	HV-400 · Stack damper (HIC-400 → 100 %)	IGN · Igniter / trial for ignition	BMS · BMS state	1ST · First-out annunciation	AMB · Flashing lamp (amber)	RED · Flashing lamp (red)	AUD · Audible alarm			
Trip Action								C	C	O	C	FO	FO	OFF	TP/RS	F	A	A	A	Notes: 1. Trip action codes: C = close; O = open; FO = fully open (controller forced to MAN 100 %); OFF = de-energise; TP = BMS tripped (purge and manual reset required); RS = BMS returns to IDLE (purge again); F = first-out; A = activate. 2. Armed only when firing (BMS in RUN or during the trial for ignition); bypassed by the BMS sequence at start-up, not by the operator. 3. Trial for ignition: the flame must be proven by BS-005 within 10 s of opening XV-301/302, otherwise the BMS trips (first-out BSLL-005 on trial). 4. Deviation alarm about the TIC-030 set point (350 °C): low 344 °C, high 356 °C. 5. Manual trip; immediate, no delay. Same effects as any SIS trip. 6. FT-008 serves FIC-003, FIAL-008 and FSLL-008 (HAZOP / Safety-layers gap: add FT-008B for the trip — HOLD). 7. BMS sequence interlock, not a trip: no fuel valves move because they are already closed; the operator restarts the purge. 8. Time delays are applied in the logic solver; trip effects latch until reset. Reset only from the BMS when no trip cause remains, then a 300 s purge. 9. Feed pumps P-100A/B are not tripped: the coil keeps its coolant flow after a trip (FV-010 fails open).		
Item No.	Tag No.	Tag Service Description	Trip/Alarm-Direction	Setting	Eng. Unit	Delay	Note no.															
001	HS-001	Emergency shutdown push-button (BMS panel / DCS)	Tripped (opened contact)	—	—	0 s	5	X	X	X	X	X	X	X	TP	X		X	X			
002	FIAL-008	Feed flow to coil low alarm (FT-008)	Low	5.0	L/s		6										X		X			
003	FSLL-008	Feed flow to coil low-low trip (FT-008)	Low-Low	3.4	L/s	5 s	6	X	X	X	X	X	X	X	TP	X		X	X			
004	TI-010	Feed after E-100 low alarm	Low	170	°C												X		X			
005	TIC-030	Heater outlet deviation alarm low (TT-040)	Low (deviation)	344	°C		4										X		X			
006	TIC-030	Heater outlet deviation alarm high (TT-040)	High (deviation)	356	°C		4										X		X			
007	TIAH-025	Heater outlet high alarm (independent TE-025)	High	360	°C												X		X			
008	TSHH-025	Heater outlet high-high trip (TE-025)	High-High	375	°C	5 s		X	X	X	X	X	X	X	TP	X		X	X			
009	TIAH-023	Firebox temperature high alarm	High	1100	°C												X		X			
010	TI-024	Radiant tube skin temperature high alarm	High	430	°C												X		X			
011	TI-021	Stack temperature high alarm	High	470	°C												X		X			
012	PIAL-020	Fuel pressure at burner gun low alarm (PT-020)	Low	100	kPa g												X		X			
013	PIAL-020	Fuel pressure at burner gun high alarm (PT-020)	High	1400	kPa g												X		X			
014	PSLL-020	Fuel pressure at burner gun low-low trip (PT-020)	Low-Low	40	kPa g	3 s	2	X	X	X	X	X	X	X	TP	X		X	X			
015	PIAL-030	Windbox draft low alarm (PT-030)	Low	1.5	mmH ₂ O												X		X			
016	PSLL-030	Windbox draft low-low trip (PT-030)	Low-Low	0.5	mmH ₂ O	3 s	2	X	X	X	X	X	X	X	TP	X		X	X			
017	PI-003	Firebox pressure high alarm (arch)	High	-0.5	mmH ₂ O												X		X			
018	AI-028	Firebox O ₂ low alarm	Low	1.0	% O ₂												X		X			
019	AI-028	Firebox O ₂ high alarm	High	6.0	% O ₂												X		X			
020	AI-029	Stack O ₂ low alarm	Low	1.5	% O ₂												X		X			
021	AI-029	Stack O ₂ high alarm	High	7.0	% O ₂												X		X			
022	AI-026	Flue gas CO high alarm	High	200	ppm												X		X			
023	BSLL-005	Flame failure (scanner BS-005) while firing	Flame lost	No flame	—	2 s	2	X	X	X	X	X	X	X	TP	X		X	X			
024	BSLL-005	Flame not proven at end of trial for ignition	Not proven	10 s trial	s	0 s	3	X	X	X	X	X	X	X	TP	X		X	X			
025	BMS-PRG	Purge permissive lost during purge	Lost	—	—		7								RS		X		X			
026	BMS-PRG	Purge credit expired (no light-off within 600 s)	Timer	600	s		7								RS		X					
027																						
028																						